Noise Cancelling Headsets – General Data

1. General

David Clark (H10–13XP) noise cancelling headset connections are available to the pilot and co-pilot for the aircraft audio integration through a David Clark Electronic Noise Cancelling (ENC) headset module (40594G–01) mounted on the upper outboard corners of the forward face of the sloping bulkhead at station 111.00.

2. Description

A. Electronic Noise Cancelling (ENC) Headset Module (40594G–01)

There are no speakers in the flight compartment. Critical aural warnings are delivered to the crew via the headsets, a functional headset is obligatory at each crew station. Standard jacks to accommodate use of a non-noise cancelling headset, in the event of failure of the noise cancelling headset, are provided at each crew station. The jacks are positioned such that it is not possible to install both a noise cancelling and a non-noise cancelling headset at the same time.

28 Volt DC power is protected by a 2 amp HEADSETS (G6) (Right Bus) circuit breaker located on the avionics circuit breaker panel (refer to 24-00-00, Electrical Power – General Data). The headset microphone and ear speakers will still function if the circuit breaker is out. The circuit breaker only provides power for the noise cancellation feature of the pilot and co-pilot headset.

The electronic noise cancelling headset module interfaces to the following aircraft system:

1) Audio Control Panel System (refer to 23-53-00, Audio Control Panel – General Data).

3. Adjustment/Test

A. Function Test

The David Clark Noise Cancelling headset module (40594G–01) can be tested with the connection of a David Clark H10–13XP headset and verifying the reduction of ambient noise when in the “on” position.

4. Removal/Installation

A. Removal

1) Ensure the aircraft electrical system is OFF. (Refer to 24-00-00, Electrical Power – General Data).

2) Open HEADSETS (G6) circuit breaker on the avionics circuit breaker panel and fit safety clip.

3) Disconnect the electrical connector from the cockpit headset box assembly.

4) Remove the screws four securing the headset box assembly unit to the bulkhead.

5) Remove the screws four securing the ENC headset module faceplate to the headset box assembly.

6) Remove required wire connections from J337 connector. Refer to the applicable wiring diagrams in Wiring Manual PSM 1-64-2W.

7) Remove the unit from aircraft.

Note Stow connectors as required.

B. Installation

1) Ensure the aircraft electrical system is OFF. (Refer to 24-00-00, Electrical Power – General Data).

2) Ensure HEADSETS (G6) circuit breaker on the avionics circuit breaker panel is opened and fitted with safety clip.

3) Install required wire connections to J337 connector. Refer to the applicable wiring diagrams in Wiring Manual PSM 1-64-2W.

4) Install the screws four securing the ENC headset module faceplate to the cockpit headset box assembly.

5) Install the screws four securing the headset box assembly unit to the bulkhead.

6) Reconnect the electrical connector to the cockpit headset box assembly. Ensure connections are secure.

7) Remove safety clip and close HEADSETS (G6) circuit breaker on the avionics circuit breaker panel.

8) Test the noise cancelling headset as per Para 3.A.

5. Troubleshooting

A. Electronic Noise Cancelling (ENC) Headset Module (45094G–01)

1) If the Electrical Noise Cancelling (ENC) Headset Module (45094G–01) is deemed inoperative check the following:

a) Ensure HEADSETS (G6) circuit breaker is closed.

b) Visually inspect the ENC module unit for damaged wires or contacts.

c) Ensure that the headset assembly connector is properly mated and secure.

d) If troubleshooting at the wire level is required refer to the applicable wiring diagrams in Wiring Manual PSM 1-64-2W.

e) If repair or service is required, send to appropriate rated avionics repair facility.